

# *Srinivasa Ramanujan*

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**Srinivasa Ramanujan** FRS (/ˈʃriːnivɑːsə rɑːˈmɑːnʊdʒən/<sup>[1]</sup> • listen (help · info); 22 December 1887 – 26 April 1920)<sup>[2]</sup> was an Indian mathematician who lived during the British Rule in India. Though he had almost no formal training in pure mathematics, he made substantial contributions to mathematical analysis, number theory, infinite series, and continued fractions, including solutions

to mathematical problems considered to be unsolvable. Ramanujan initially developed his own mathematical research in isolation: "He tried to interest the leading professional mathematicians in his work, but failed for the most part. What he had to show them was too novel, too unfamiliar, and additionally presented in unusual ways; they could not be bothered".<sup>[3]</sup> Seeking mathematicians who could better understand his work, in 1913 he began a postal partnership with the English mathematician G. H. Hardy at the University of Cambridge, England. Recognizing the extraordinary work sent to him as samples, Hardy arranged travel

for Ramanujan to Cambridge. In his notes, Ramanujan had produced groundbreaking new theorems, including some that Hardy stated had "defeated [him and his colleagues] completely", in addition to rediscovering recently proven but highly advanced results.

# Srinivasa Ramanujan

FRS



**Born**

22 December 1887

Erode, Madras

Presidency, British

India (present-day

Tamil Nadu, India)

**Died**

26 April 1920

(aged 32)

Kumbakonam, Madras

Presidency, British

India (present-day

Tamil Nadu, India)

## Residence

- Kumbakonam,  
Madras Presidency,  
British India  
(present-day Tamil Nadu, India)
- Madras, Madras Presidency, British India (present-day Chennai, Tamil Nadu, India)
- London, England,  
United Kingdom of Great Britain and Ireland (present-day United Kingdom)

## Nationality

Indian

## Education

Government Arts College (no degree)

Pachaivanna's College

Trinity College

(no degree)

Trinity College,

Cambridge (BSc, 1916)

## Known for

Landau–Ramanujan  
constant

Mock theta functions

Ramanujan conjecture

Ramanujan prime

Ramanujan–Soldner  
constant

Ramanujan theta  
function

Ramanujan's sum

Rogers–Ramanujan  
identities

Ramanujan's master  
theorem

Ramanujan–Sato

|                                                                                     |                                                                |
|-------------------------------------------------------------------------------------|----------------------------------------------------------------|
|                                                                                     | <u>series</u>                                                  |
| <b>Awards</b>                                                                       | <u>Fellow of the Royal Society.</u>                            |
| <b>Scientific career</b>                                                            |                                                                |
| <b>Fields</b>                                                                       | <u>Mathematics</u>                                             |
| <b>Institutions</b>                                                                 | <u>Trinity College,</u><br><u>Cambridge</u>                    |
| <b><u>Thesis</u></b>                                                                | <i><u>Highly Composite</u></i><br><i><u>Numbers</u></i> (1916) |
| <b>Academic advisors</b>                                                            | <u>G. H. Hardy.</u><br><u>J. E. Littlewood</u>                 |
| <b>Influences</b>                                                                   | <u>G. S. Carr</u>                                              |
| <b>Influenced</b>                                                                   | <u>G. H. Hardy.</u>                                            |
| <b>Signature</b>                                                                    |                                                                |
|  |                                                                |

During his short life, Ramanujan independently compiled nearly 3,900 results (mostly identities and equations).<sup>[4]</sup> Many were completely novel; his original and highly unconventional results, such as the Ramanujan prime, the Ramanujan theta function, partition formulae and mock theta functions, have opened entire new areas of work and inspired a vast amount of further research.<sup>[5]</sup> Nearly all his claims have now been proven correct.<sup>[6]</sup> *The Ramanujan Journal*, a peer-reviewed scientific journal, was established to publish work in all areas of mathematics influenced by Ramanujan,<sup>[7]</sup> and his notebooks—containing summaries of his

published and unpublished results—have been analyzed and studied for decades since his death as a source of new mathematical ideas. As late as 2011 and again in 2012, researchers continued to discover that mere comments in his writings about "simple properties" and "similar outputs" for certain findings were themselves profound and subtle number theory results that remained unsuspected until nearly a century after his death.<sup>[8][9]</sup> He became one of the youngest Fellows of the Royal Society and only the second Indian member, and the first Indian to be elected a Fellow of Trinity College, Cambridge. Of his original letters, Hardy stated that a single look

was enough to show they could only have been written by a mathematician of the highest calibre, comparing Ramanujan to other mathematical geniuses such as Euler and Jacobi.

In 1919, ill health—now believed to have been hepatic amoebiasis (a complication from episodes of dysentery, many years previously)—compelled Ramanujan's return to India, where he died in 1920 at the age of 32. His last letters to Hardy, written January 1920, show that he was still continuing to produce new mathematical ideas and theorems. His "lost notebook", containing discoveries from the last year of his life, caused great

excitement among mathematicians when it was rediscovered in 1976.

A deeply religious Hindu,<sup>[10]</sup> Ramanujan credited his substantial mathematical capacities to divinity, and stated that the mathematical knowledge he displayed was revealed to him by his family goddess. "An equation for me has no meaning," he once said, "unless it expresses a thought of God."<sup>[11]</sup>

## Early life

*Ramanujan's birthplace on 18 Alahiri Street, Erode*

*Ramanujan's home on Sarangapani Sannidhi Street,  
Kumbakonam*

Ramanujan (literally, "younger brother of Rama", a Hindu deity<sup>[12]:12</sup>) was born on 22 December 1887 into a Tamil Brahmin

Iyengar family in Erode, Madras  
Presidency. (now Tamil Nadu), at the  
residence of his maternal  
grandparents.<sup>[12]:11</sup> His father,  
Kuppuswamy Srinivasa Iyengar, originally  
from Thanjavur district, worked as a clerk  
in a sari shop.<sup>[12]:17–18</sup><sup>[13]</sup> His mother,  
Komalatammal, was a housewife and  
also sang at a local temple.<sup>[14]</sup> They lived  
in a small traditional home on  
Sarangapani Sannidhi Street in the town  
of Kumbakonam.<sup>[15]</sup> The family home is  
now a museum. When Ramanujan was a  
year and a half old, his mother gave birth  
to a son, Sadagopan, who died less than  
three months later. In December 1889,  
Ramanujan contracted smallpox, though

he recovered, unlike 4,000 others who would die in a bad year in the Thanjavur district around this time. He moved with his mother to her parents' house in Kanchipuram, near Madras (now Chennai). His mother gave birth to two more children, in 1891 and 1894, both failing to reach their first birthdays.<sup>[12]:12</sup>

On 1 October 1892, Ramanujan was enrolled at the local school.<sup>[12]:13</sup> After his maternal grandfather lost his job as a court official in Kanchipuram,<sup>[12]:19</sup>

Ramanujan and his mother moved back to Kumbakonam and he was enrolled in the Kangayan Primary School.<sup>[12]:14</sup> When his paternal grandfather died, he was

sent back to his maternal grandparents, then living in Madras. He did not like school in Madras, and tried to avoid attending. His family enlisted a local constable to make sure the boy attended school. Within six months, Ramanujan was back in Kumbakonam.<sup>[12]:14</sup>

Since Ramanujan's father was at work most of the day, his mother took care of the boy as a child. He had a close relationship with her. From her, he learned about tradition and puranas. He learned to sing religious songs, to attend pujas at the temple, and to maintain particular eating habits—all of which are part of Brahmin culture.<sup>[12]:20</sup> At the

Kangayan Primary School, Ramanujan performed well. Just before turning 10, in November 1897, he passed his primary examinations in English, Tamil, geography and arithmetic with the best scores in the district.<sup>[12]:25</sup> That year, Ramanujan entered Town Higher Secondary School, where he encountered formal mathematics for the first time.<sup>[12]:25</sup>

By age 11, he had exhausted the mathematical knowledge of two college students who were lodgers at his home. He was later lent a book by S. L. Loney on advanced trigonometry.<sup>[16][17]</sup> He mastered this by the age of 13 while

discovering sophisticated theorems on his own. By 14, he was receiving merit certificates and academic awards that continued throughout his school career, and he assisted the school in the logistics of assigning its 1200 students (each with differing needs) to its approximately 35 teachers.<sup>[12]:27</sup> He completed mathematical exams in half the allotted time, and showed a familiarity with geometry and infinite series. Ramanujan was shown how to solve cubic equations in 1902; he developed his own method to solve the quartic. The following year, Ramanujan tried to solve the quintic, not knowing that it could not be solved by radicals.

In 1903, when he was 16, Ramanujan obtained from a friend a library copy of A Synopsis of Elementary Results in Pure and Applied Mathematics, G. S. Carr's collection of 5,000 theorems.<sup>[12]:39[18]</sup>

Ramanujan reportedly studied the contents of the book in detail.<sup>[19]</sup> The book is generally acknowledged as a key element in awakening his genius.<sup>[19]</sup> The next year, Ramanujan independently developed and investigated the Bernoulli numbers and calculated the Euler–Mascheroni constant up to 15 decimal places.<sup>[12]:90</sup> His peers at the time commented that they "rarely understood him" and "stood in respectful awe" of him.<sup>[12]:27</sup>

When he graduated from Town Higher Secondary School in 1904, Ramanujan was awarded the K. Ranganatha Rao prize for mathematics by the school's headmaster, Krishnaswami Iyer. Iyer introduced Ramanujan as an outstanding student who deserved scores higher than the maximum.<sup>[12]</sup> He received a scholarship to study at Government Arts College, Kumbakonam,<sup>[12]:28[12]:45</sup> but was so intent on mathematics that he could not focus on any other subjects and failed most of them, losing his scholarship in the process.<sup>[12]:47</sup> In August 1905, Ramanujan ran away from home, heading towards Visakhapatnam, and stayed in Rajahmundry.<sup>[20]</sup> for about a

month.<sup>[12]:47–48</sup> He later enrolled at Pachaiyappa's College in Madras. There he passed in mathematics, choosing only to attempt questions that appealed to him and leaving the rest unanswered, but performed poorly in other subjects, such as English, physiology and Sanskrit.<sup>[21]</sup> Ramanujan failed his Fellow of Arts exam in December 1906 and again a year later. Without a FA degree, he left college and continued to pursue independent research in mathematics, living in extreme poverty and often on the brink of starvation.<sup>[12]:55–56</sup>

It was in 1910, after a meeting between the 23-year-old Ramanujan and the

founder of the Indian Mathematical Society, V. Ramaswamy Aiyer, also known as Professor Ramaswami, that Ramanujan started to get recognition within the mathematics circles of Madras, subsequently leading to his inclusion as a researcher at the University of Madras.<sup>[22]</sup>

## Adulthood in India

On 14 July 1909, Ramanujan married Janaki (Janakiammal) (21 March 1899 – 13 April 1994),<sup>[23]</sup> a girl whom his mother had selected for him a year earlier and who was ten years old when they married.<sup>[24][25][12]:71</sup> It was not unusual for

marriages to be arranged with girls. She came from Rajendram, a village close to Marudur (Karur district) Railway Station. Ramanujan's father did not participate in the marriage ceremony.<sup>[26]</sup> As was common at that time, Janakiammal continued to stay at her maternal home for three years after marriage till she attained puberty. In 1912, she and Ramanujan's mother joined Ramanujan in Madras.<sup>[27]</sup>

After the marriage, Ramanujan developed a hydrocele testis.<sup>[12]:72</sup> The condition could be treated with a routine surgical operation that would release the blocked fluid in the scrotal sac, but his family did

not have the money for the operation. In January 1910, a doctor volunteered to do the surgery at no cost.<sup>[28]</sup>

After his successful surgery, Ramanujan searched for a job. He stayed at a friend's house while he went from door to door around Madras looking for a clerical position. To make money, he tutored students at Presidency College who were preparing for their F.A. exam.<sup>[12]:73</sup>

In late 1910, Ramanujan was sick again. He feared for his health, and told his friend R. Radakrishna Iyer to "hand [his notebooks] over to Professor Singaravelu Mudaliar [the mathematics professor at Pachaiyappa's College] or to the British

professor Edward B. Ross, of the Madras Christian College.<sup>[12]:74–75</sup> After

Ramanujan recovered and retrieved his notebooks from Iyer, he took a train from Kumbakonam to Villupuram, a city under French control.<sup>[29][30]</sup> In 1912, Ramanujan moved to a house in Saiva Muthaiah Mudali street, George Town, Madras with his wife and mother where they lived for a few months.<sup>[31]</sup> In May 1913, upon securing a research position at Madras University, Ramanujan moved with his family to Triplicane.<sup>[32]</sup>

## **Pursuit of career in mathematics**

Ramanujan met deputy collector V. Ramaswamy Aiyer, who had founded the Indian Mathematical Society.<sup>[12]</sup>:77  
Wishing for a job at the revenue department where Aiyer worked, Ramanujan showed him his mathematics notebooks. As Aiyer later recalled:

*I was struck by the extraordinary mathematical results contained in [the notebooks]. I had no mind to smother his genius by an appointment in the lowest rungs of the revenue department.*<sup>[33]</sup>

|

Aiyer sent Ramanujan, with letters of introduction, to his mathematician friends in Madras.<sup>[12]:77</sup> Some of them looked at his work and gave him letters of introduction to R. Ramachandra Rao, the district collector for Nellore and the secretary of the Indian Mathematical Society.<sup>[34][35][36]</sup> Rao was impressed by Ramanujan's research but doubted that it was his own work. Ramanujan mentioned a correspondence he had with Professor Saldhana, a notable Bombay mathematician, in which Saldhana expressed a lack of understanding of his work but concluded that he was not a phony.<sup>[12]:80</sup> Ramanujan's friend C. V.

Rajagopalachari tried to quell Rao's doubts about Ramanujan's academic integrity. Rao agreed to give him another chance, and listened as Ramanujan discussed elliptic integrals, hypergeometric series, and his theory of divergent series, which Rao said ultimately converted him to a belief in Ramanujan's brilliance.<sup>[12]:80</sup> When Rao asked him what he wanted, Ramanujan replied that he needed work and financial support. Rao consented and sent him to Madras. He continued his research, with Rao's financial aid taking care of his daily needs. With Aiyer's help, Ramanujan had his work published in the *Journal of the Indian Mathematical Society*.<sup>[12]:86</sup>

One of the first problems he posed in the journal was to find the value of:

He waited for a solution to be offered in three issues, over six months, but failed to receive any. At the end, Ramanujan supplied the solution to the problem himself. On page 105 of his first notebook, he formulated an equation that could be used to solve the infinitely nested radicals problem.

Using this equation, the answer to the question posed in the *Journal* was simply 3, obtained by setting  $x = 2$ ,  $n = 1$ , and  $a = 0$ .<sup>[12]:87</sup> Ramanujan wrote his first formal paper for the *Journal* on the properties of Bernoulli numbers. One property he discovered was that the denominators (sequence A027642 in the OEIS) of the fractions of Bernoulli numbers were always divisible by six. He also devised a method of calculating  $B_n$  based on previous Bernoulli numbers. One of these methods follows:

It will be observed that if  $n$  is even but not equal to zero,

1.  $B_n$  is a fraction and the numerator of  $\frac{B_n}{n}$  in its lowest terms is a prime number,
2. the denominator of  $B_n$  contains each of the factors 2 and 3 once and only once,
3.  $2^n(2^n - 1)\frac{B_n}{n}$  is an integer and  $2(2^n - 1)B_n$  consequently is an *odd* integer.

In his 17-page paper, "Some Properties of Bernoulli's Numbers" (1911), Ramanujan gave three proofs, two corollaries and three conjectures.<sup>[12]:91</sup> Ramanujan's writing initially had many flaws. As *Journal* editor M. T. Narayana Iyengar noted:

*Mr. Ramanujan's methods were so terse and novel and his presentation so lacking in clearness and precision, that the ordinary [mathematical reader], unaccustomed to such intellectual gymnastics, could hardly follow him.*<sup>[37]</sup>

Ramanujan later wrote another paper and also continued to provide problems in the *Journal*.<sup>[38]</sup> In early 1912, he got a temporary job in the Madras Accountant General's office, with a salary of 20 rupees per month. He lasted only a few weeks.<sup>[39]</sup> Toward the end of that

assignment, he applied for a position under the Chief Accountant of the Madras Port Trust.

In a letter dated 9 February 1912, Ramanujan wrote:

*Sir,*

*I understand there is a clerkship vacant in your office, and I beg to apply for the same. I have passed the Matriculation Examination and studied up to the F.A. but was prevented from pursuing my studies further owing to*

*several untoward  
circumstances. I have,  
however, been devoting all my  
time to Mathematics and  
developing the subject. I can  
say I am quite confident I can  
do justice to my work if I am  
appointed to the post. I  
therefore beg to request that  
you will be good enough to  
confer the appointment on  
me.*<sup>[40]</sup>

Attached to his application was a  
recommendation from E. W. Middlemast,  
a mathematics professor at the

Presidency College, who wrote that Ramanujan was "a young man of quite exceptional capacity in Mathematics".<sup>[41]</sup> Three weeks after he had applied, on 1 March, Ramanujan learned that he had been accepted as a Class III, Grade IV accounting clerk, making 30 rupees per month.<sup>[12]:96</sup> At his office, Ramanujan easily and quickly completed the work he was given, so he spent his spare time doing mathematical research.

Ramanujan's boss, Sir Francis Spring, and S. Narayana Iyer, a colleague who was also treasurer of the Indian Mathematical Society, encouraged Ramanujan in his mathematical pursuits.

## Contacting British mathematicians

In the spring of 1913, Narayana Iyer, Ramachandra Rao and E. W. Middlemast tried to present Ramanujan's work to British mathematicians. M. J. M. Hill of University College London commented that Ramanujan's papers were riddled with holes.<sup>[12]:105</sup> He said that although Ramanujan had "a taste for mathematics, and some ability," he lacked the educational background and foundation needed to be accepted by mathematicians.<sup>[42]</sup> Although Hill did not offer to take Ramanujan on as a student, he did give thorough and serious

professional advice on his work. With the help of friends, Ramanujan drafted letters to leading mathematicians at Cambridge University.<sup>[12]:106</sup>

The first two professors, H. F. Baker and E. W. Hobson, returned Ramanujan's papers without comment.<sup>[12]:170–171</sup> On 16 January 1913, Ramanujan wrote to G. H. Hardy. Coming from an unknown mathematician, the nine pages of mathematics made Hardy initially view Ramanujan's manuscripts as a possible fraud.<sup>[43]</sup> Hardy recognised some of Ramanujan's formulae but others "seemed scarcely possible to believe".<sup>[44]:494</sup> One of the theorems

Hardy found amazing was on the bottom of page three (valid for  $0 < a < b + \frac{1}{2}$ ):

Hardy was also impressed by some of Ramanujan's other work relating to infinite series:

The first result had already been determined by G. Bauer in 1859. The second was new to Hardy, and was derived from a class of functions called hypergeometric series, which had first been researched by Leonhard Euler and Carl Friedrich Gauss. Hardy found these results "much more intriguing" than Gauss's work on integrals.<sup>[12]:167</sup> After seeing Ramanujan's theorems on continued fractions on the last page of the manuscripts, Hardy commented that the theorems "defeated me completely; I had never seen anything in the least like them before".<sup>[12]:168</sup> He figured that Ramanujan's theorems "must be true, because, if they were not true, no one

would have the imagination to invent them".<sup>[12]:168</sup> Hardy asked a colleague, J. E. Littlewood, to take a look at the papers. Littlewood was amazed by Ramanujan's genius. After discussing the papers with Littlewood, Hardy concluded that the letters were "certainly the most remarkable I have received" and said that Ramanujan was "a mathematician of the highest quality, a man of altogether exceptional originality and power".<sup>[44]:494–495</sup> One colleague, E. H. Neville, later remarked that "not one [theorem] could have been set in the most advanced mathematical examination in the world".<sup>[38]</sup>

On 8 February 1913, Hardy wrote Ramanujan a letter expressing his interest in his work, adding that it was "essential that I should see proofs of some of your assertions".<sup>[45]</sup> Before his letter arrived in Madras during the third week of February, Hardy contacted the Indian Office to plan for Ramanujan's trip to Cambridge. Secretary Arthur Davies of the Advisory Committee for Indian Students met with Ramanujan to discuss the overseas trip.<sup>[46]</sup> In accordance with his Brahmin upbringing, Ramanujan refused to leave his country to "go to a foreign land".<sup>[12]:185</sup> Meanwhile, he sent Hardy a letter packed with theorems,

writing, "I have found a friend in you who views my labour sympathetically."<sup>[47]</sup>

To supplement Hardy's endorsement, Gilbert Walker, a former mathematical lecturer at Trinity College, Cambridge, looked at Ramanujan's work and expressed amazement, urging the young man to spend time at Cambridge.<sup>[12]:175</sup>

As a result of Walker's endorsement, B. Hanumantha Rao, a mathematics professor at an engineering college, invited Ramanujan's colleague Narayana Iyer to a meeting of the Board of Studies in Mathematics to discuss "what we can do for S. Ramanujan".<sup>[48]</sup> The board agreed to grant Ramanujan a research

scholarship of 75 rupees per month for the next two years at the University of Madras.<sup>[49]</sup> While he was engaged as a research student, Ramanujan continued to submit papers to the *Journal of the Indian Mathematical Society*. In one instance, Narayana Iyer submitted some of Ramanujan's theorems on summation of series to the journal, adding, "The following theorem is due to S.

Ramanujan, the mathematics student of Madras University." Later in November, British Professor Edward B. Ross of Madras Christian College, whom Ramanujan had met a few years before, stormed into his class one day with his eyes glowing, asking his students, "Does

Ramanujan know Polish?" The reason was that in one paper, Ramanujan had anticipated the work of a Polish mathematician whose paper had just arrived in the day's mail.<sup>[50]</sup> In his quarterly papers, Ramanujan drew up theorems to make definite integrals more easily solvable. Working off Giuliano Frullani's 1821 integral theorem, Ramanujan formulated generalisations that could be made to evaluate formerly unyielding integrals.<sup>[12]:183</sup>

Hardy's correspondence with Ramanujan soured after Ramanujan refused to come to England. Hardy enlisted a colleague lecturing in Madras, E. H. Neville, to

mentor and bring Ramanujan to England.<sup>[12]:184</sup> Neville asked Ramanujan why he would not go to Cambridge. Ramanujan apparently had now accepted the proposal; as Neville put it, "Ramanujan needed no converting and that his parents' opposition had been withdrawn".<sup>[38]</sup> Apparently, Ramanujan's mother had a vivid dream in which the family goddess, the deity of Namagiri, commanded her "to stand no longer between her son and the fulfilment of his life's purpose".<sup>[38]</sup> Ramanujan traveled to England by ship, leaving his wife to stay with his parents in India.

## Life in England

*Ramanujan (centre) and his colleague G. H. Hardy (extreme right), with other scientists, outside the Senate House, Cambridge, c.1914–19*

*Whewell's Court, Trinity College, Cambridge*

Ramanujan departed from Madras aboard the S.S. *Nevasa* on 17 March 1914.<sup>[12]:196</sup> When he disembarked in London on 14 April, Neville was waiting for him with a car. Four days later, Neville took him to his house on Chesterton Road in Cambridge. Ramanujan immediately began his work with Littlewood and Hardy. After six weeks, Ramanujan moved out of Neville's house and took up residence on Whewell's Court, a five-minute walk from Hardy's room.<sup>[12]:202</sup> Hardy and Littlewood began to look at Ramanujan's notebooks. Hardy had already received 120 theorems from Ramanujan in the first two letters, but there were many more results and

theorems in the notebooks. Hardy saw that some were wrong, others had already been discovered, and the rest were new breakthroughs.<sup>[51]</sup> Ramanujan left a deep impression on Hardy and Littlewood. Littlewood commented, "I can believe that he's at least a Jacobi",<sup>[52]</sup> while Hardy said he "can compare him only with Euler or Jacobi."<sup>[53]</sup>

Ramanujan spent nearly five years in Cambridge collaborating with Hardy and Littlewood, and published part of his findings there. Hardy and Ramanujan had highly contrasting personalities. Their collaboration was a clash of different cultures, beliefs, and working styles. In

the previous few decades, the foundations of mathematics had come into question and the need for mathematically rigorous proofs recognized. Hardy was an atheist and an apostle of proof and mathematical rigour, whereas Ramanujan was a deeply religious man who relied very strongly on his intuition and insights. While in England, Hardy tried his best to fill the gaps in Ramanujan's education and to mentor him in the need for formal proofs to support his results, without hindering his inspiration—a conflict that neither found easy.

Ramanujan was awarded a Bachelor of Science degree by research (this degree was later renamed PhD) in March 1916 for his work on highly composite numbers, the first part of which was published as a paper in the *Proceedings of the London Mathematical Society*. The paper was more than 50 pages and proved various properties of such numbers. Hardy remarked that it was one of the most unusual papers seen in mathematical research at that time and that Ramanujan showed extraordinary ingenuity in handling it. On 6 December 1917, he was elected to the London Mathematical Society. In 1918 he was elected a Fellow of the Royal Society, the

second Indian admitted to the Royal Society, following Ardaseer Cursetjee in 1841. At age 31 Ramanujan was one of the youngest Fellows in the history of the Royal Society. He was elected "for his investigation in Elliptic functions and the Theory of Numbers." On 13 October 1918, he was the first Indian to be elected a Fellow of Trinity College, Cambridge.<sup>[12]</sup>:299–300

## **Illness and death**

Throughout his life, Ramanujan was plagued by health problems. His health worsened in England; possibly he was also less resilient due to the difficulty of

keeping to the strict dietary requirements of his religion in England and wartime rationing during 1914–1918. He was diagnosed with tuberculosis and a severe vitamin deficiency at the time, and was confined to a sanatorium. In 1919 he returned to Kumbakonam, Madras Presidency, and soon thereafter, in 1920, died at the age of 32. After his death, his brother Tirunarayanan chronicled Ramanujan's remaining handwritten notes consisting of formulae on singular moduli, hypergeometric series and continued fractions and compiled them.<sup>[27]</sup>

Ramanujan's widow, Smt. Janaki Ammal, moved to Bombay; in 1931 she returned to Madras and settled in Triplicane, where she supported herself on a pension from Madras University and income from tailoring. In 1950, she adopted a son, W. Narayanan, who eventually became an officer of the State Bank of India and raised a family. In her later years, she was granted a lifetime pension from Ramanujan's former employer, the Madras Port Trust, and was also granted pensions from, among others, the Indian National Science Academy and the state governments of Tamil Nadu, Andhra Pradesh and West Bengal. She continued to cherish

Ramanujan's memory, and was active in efforts towards increasing his public recognition; prominent mathematicians, including George Andrews, Bruce C. Berndt and Béla Bollobás made it a point to visit her while in India. She died at her Triplicane residence in 1994.<sup>[26][27]</sup>

A 1994 analysis of Ramanujan's medical records and symptoms by Dr. D. A. B. Young<sup>[54]</sup> concluded that his medical symptoms—including his past relapses, fevers, and hepatic conditions—were much closer to those resulting from hepatic amoebiasis, an illness then widespread in Madras, rather than tuberculosis. He had two episodes of

dysentery before he left India. When not properly treated, dysentery can lie dormant for years and lead to hepatic amoebiasis, whose diagnosis was not then well established.<sup>[55]</sup> At the time, if properly diagnosed, amoebiasis was a treatable and often curable disease;<sup>[55][56]</sup> for instance, British soldiers who had contracted the disease during the First World War were being successfully cured of amoebiasis around the time Ramanujan left England.<sup>[57]</sup>

## **Personality and spiritual life**

Ramanujan has been described as a person of a somewhat shy and quiet

disposition, a dignified man with pleasant manners.<sup>[58]</sup> He lived a simple life at Cambridge.<sup>[12]:234,241</sup> Ramanujan's first Indian biographers describe him as a rigorously orthodox Hindu. He credited his acumen to his family\_goddess, Namagiri Thayar (Goddess Mahalakshmi) of Namakkal. He looked to her for inspiration in his work<sup>[12]:36</sup> and said he dreamed of blood drops that symbolised her consort, Narasimha. Afterward he would receive visions of scrolls of complex mathematical content unfolding before his eyes.<sup>[12]:281</sup> He often said, "An equation for me has no meaning unless it represents a thought of God."<sup>[59]</sup>

Hardy cites Ramanujan as remarking that all religions seemed equally true to him.<sup>[12]:283</sup> Hardy further argued that Ramanujan's religious belief had been romanticised by Westerners and overstated—in reference to his belief, not practice—by Indian biographers. At the same time, he remarked on Ramanujan's strict vegetarianism.<sup>[60]</sup>

## Mathematical achievements

In mathematics, there is a distinction between insight and formulating or working through a proof. Ramanujan proposed an abundance of formulae that could be investigated later in depth. G. H.

Hardy said that Ramanujan's discoveries are unusually rich and that there is often more to them than initially meets the eye. As a byproduct of his work, new directions of research were opened up. Examples of the most interesting of these formulae include the intriguing infinite series for  $\pi$ , one of which is given below:

This result is based on the negative fundamental discriminant

$d = -4 \times 58 = -232$  with class number

$h(d) = 2$ . Further,  $26390 = 5 \times 7 \times 13 \times 58$

and  $16 \times 9801 = 396^2$  and is related to the fact that

This might be compared to Heegner numbers, which have class number 1 and yield similar formulae.

Ramanujan's series for  $\pi$  converges extraordinarily rapidly (exponentially) and forms the basis of some of the fastest algorithms currently used to calculate  $\pi$ . Truncating the sum to the first term also gives the approximation  $\frac{9801\sqrt{2}}{4412}$  for  $\pi$ , which is correct to six decimal places; truncating it to the first two terms gives a value correct to 14 decimal places. See

also the more general Ramanujan–Sato series.

One of Ramanujan's remarkable capabilities was the rapid solution of problems, illustrated by the following anecdote about an incident in which P. C. Mahalanobis posed a problem:

*Imagine that you are on a street with houses marked 1 through  $n$ . There is a house in between ( $x$ ) such that the sum of the house numbers to the left of it equals the sum of the house numbers to its right. If  $n$*

is between 50 and 500, what are  $n$  and  $x$ ?' This is a bivariate problem with multiple solutions. Ramanujan thought about it and gave the answer with a twist: He gave a continued fraction. The unusual part was that it was the solution to the whole class of problems. Mahalanobis was astounded and asked how he did it. 'It is simple. The minute I heard the problem, I knew that the answer was a continued fraction. Which continued

*fraction, I asked myself. Then the answer came to my mind', Ramanujan replied."*<sup>[61]</sup><sup>[62]</sup>

His intuition also led him to derive some previously unknown identities, such as

for all  $\theta$ , where  $\Gamma(z)$  is the gamma function, and related to a special value of the Dedekind eta function. Expanding

into series of powers and equating coefficients of  $\theta^0$ ,  $\theta^4$ , and  $\theta^8$  gives some deep identities for the hyperbolic secant.

In 1918 Hardy and Ramanujan studied the partition function  $P(n)$  extensively.

They gave a non-convergent asymptotic series that permits exact computation of the number of partitions of an integer.

Hans Rademacher, in 1937, was able to refine their formula to find an exact convergent series solution to this problem. Ramanujan and Hardy's work in this area gave rise to a powerful new method for finding asymptotic formulae called the circle method.<sup>[63]</sup>

In the last year of his life, Ramanujan discovered mock theta functions.<sup>[64]</sup> For many years these functions were a mystery, but they are now known to be the holomorphic parts of harmonic weak Maass forms.

## **The Ramanujan conjecture**

Although there are numerous statements that could have borne the name *Ramanujan conjecture*, there is one that was highly influential on later work. In particular, the connection of this conjecture with conjectures of André Weil in algebraic geometry opened up new areas of research. That Ramanujan

conjecture is an assertion on the size of the tau-function, which has as generating function the discriminant modular form  $\Delta(q)$ , a typical cusp form in the theory of modular forms. It was finally proven in 1973, as a consequence of Pierre Deligne's proof of the Weil conjectures. The reduction step involved is complicated. Deligne won a Fields Medal in 1978 for that work.<sup>[5]</sup>

In his paper "On certain arithmetical functions", Ramanujan defined the so-called delta-function whose coefficients are called  $\tau(n)$  (the Ramanujan tau function).<sup>[65]</sup> He proved many congruences for these numbers such as

$\tau(p) \equiv 1 + p^{11} \pmod{691}$  for primes  $p$ . This congruence (and others like it that Ramanujan proved) inspired Jean-Pierre Serre (1954 Fields Medalist) to conjecture that there is a theory of Galois representations which "explains" these congruences and more generally all modular forms.  $\Delta(z)$  is the first example of a modular form to be studied in this way. Pierre Deligne (in his Fields Medal-winning work) proved Serre's conjecture. The proof of Fermat's Last Theorem proceeds by first reinterpreting elliptic curves and modular forms in terms of these Galois representations. Without this theory there would be no proof of Fermat's Last Theorem.<sup>[66]</sup>

# Ramanujan's notebooks

While still in Madras, Ramanujan recorded the bulk of his results in four notebooks of loose-leaf paper. They were mostly written up without any derivations. This is probably the origin of the misapprehension that Ramanujan was unable to prove his results and simply thought up the final result directly. Mathematician Bruce C. Berndt, in his review of these notebooks and Ramanujan's work, says that Ramanujan most certainly was able to prove most of his results, but chose not to.

This may have been for any number of reasons. Since paper was very expensive, Ramanujan would do most of his work and perhaps his proofs on slate, and then transfer just the results to paper. Using a slate was common for mathematics students in the Madras Presidency at the time. He was also quite likely to have been influenced by the style of G. S. Carr's book, which stated results without proofs. Finally, it is possible that Ramanujan considered his workings to be for his personal interest alone and therefore recorded only the results.<sup>[67]</sup>

The first notebook has 351 pages with 16 somewhat organised chapters and some

unorganised material. The second notebook has 256 pages in 21 chapters and 100 unorganised pages, with the third notebook containing 33 unorganised pages. The results in his notebooks inspired numerous papers by later mathematicians trying to prove what he had found. Hardy himself created papers exploring material from Ramanujan's work, as did G. N. Watson, B. M. Wilson, and Bruce Berndt.<sup>[67]</sup> A fourth notebook with 87 unorganised pages, the so-called "lost notebook", was rediscovered in 1976 by George Andrews.<sup>[55]</sup>

## Hardy–Ramanujan number

# 1729

The number 1729 is known as the Hardy–Ramanujan number after a famous visit by Hardy to see Ramanujan at a hospital. In Hardy's words:<sup>[68]</sup>

*I remember once going to see him when he was ill at Putney. I had ridden in taxi cab number 1729 and remarked that the number seemed to me rather a dull one, and that I hoped it was not an unfavorable omen. "No", he replied, "it is a very interesting number; it is the*

*smallest number expressible as the sum of two cubes in two different ways."*

Immediately before this anecdote, Hardy quoted Littlewood as saying, "Every positive integer was one of [Ramanujan's] personal friends."<sup>[69]</sup>

The two different ways are:

Generalizations of this idea have created the notion of "taxicab numbers".

## Mathematicians' views of Ramanujan

In his obituary of Ramanujan, which he wrote for *Nature* in 1920, Hardy observed Ramanujan's work primarily involved fields less known even amongst other pure mathematicians, concluding:

*His insight into formulae was quite amazing, and altogether beyond anything I have met with in any European mathematician. It is perhaps useless to speculate as to his history had he been introduced to modern ideas and methods at sixteen instead of at twenty-six. It is not extravagant to*

*suppose that he might have become the greatest mathematician of his time. What he actually did is wonderful enough...when the researches which his work has suggested have been completed, it will probably seem a good deal more wonderful than it does to-day.*<sup>[44]</sup>

Hardy further said:

*He combined a power of generalization, a feeling for form, and a capacity for rapid modification of his hypotheses, that were often really startling, and made him, in his own peculiar field, without a rival in his day. The limitations of his knowledge were as startling as its profundity. Here was a man who could work out modular equations and theorems... to orders unheard of, whose mastery of continued fractions was... beyond that of any*

mathematician in the world,  
who had found for himself the  
functional equation of the zeta  
function and the dominant  
terms of many of the most  
famous problems in the  
analytic theory of numbers;  
and yet he had never heard of a  
doubly periodic function or of  
Cauchy's theorem, and had  
indeed but the vaguest idea of  
what a function of a complex  
variable was...".<sup>[70]</sup>

When asked about the methods Ramanujan employed to arrive at his solutions, Hardy said that they were "arrived at by a process of mingled argument, intuition, and induction, of which he was entirely unable to give any coherent account."<sup>[71]</sup> He also stated that he had "never met his equal, and can compare him only with Euler or Jacobi."<sup>[71]</sup>

K. Srinivasa Rao has said,<sup>[72]</sup> "As for his place in the world of Mathematics, we quote Bruce C. Berndt: 'Paul Erdős has passed on to us Hardy's personal ratings of mathematicians. Suppose that we rate mathematicians on the basis of pure

talent on a scale from 0 to 100, Hardy gave himself a score of 25, J. E. Littlewood 30, David Hilbert 80 and Ramanujan 100." During a lecture at IIT Madras in May 2011, Berndt stated that over the last 40 years, as nearly all of Ramanujan's theorems have been proven right, there had been greater appreciation of Ramanujan's work and brilliance, and that Ramanujan's work was now pervading many areas of modern mathematics and physics.<sup>[64][73]</sup>

In his book *Scientific Edge*, the physicist Jayant Narlikar spoke of "Srinivasa Ramanujan, discovered by the Cambridge mathematician Hardy, whose great

mathematical findings were beginning to be appreciated from 1915 to 1919. His achievements were to be fully understood much later, well after his untimely death in 1920. For example, his work on the highly composite numbers (numbers with a large number of factors) started a whole new line of investigations in the theory of such numbers."

## Posthumous recognition

*Bust of Ramanujan in the garden of Birla Industrial & Technological Museum*

*The 2012 Indian stamp dedicated to the National Mathematics Day and featuring Ramanujan*

The year after his death, Nature listed Ramanujan among other distinguished scientists and mathematicians on a "Calendar of Scientific Pioneers," who had achieved eminence.<sup>[74]</sup> Ramanujan's home state of Tamil Nadu celebrates 22 December (Ramanujan's birthday) as 'State IT Day'. Stamp picturing

Ramanujan were issued by the Government of India in 1962, 2011, 2012 and 2016.<sup>[75]</sup>

Since Ramanujan's centennial year, his birthday, 22 December, has been annually celebrated as Ramanujan Day by the Government Arts College, Kumbakonam where he studied and at the IIT Madras in Chennai. A prize for young mathematicians from developing countries has been created in Ramanujan's name by the International Centre for Theoretical Physics (ICTP) in cooperation with the International Mathematical Union, which nominate members of the prize committee.

SASTRA University, a private university based in the state of Tamil Nadu in South India, has instituted the SASTRA Ramanujan Prize of US\$10,000 to be given annually to a mathematician not exceeding the age of 32 for outstanding contributions in an area of mathematics influenced by Ramanujan. Based on the recommendations of a high level committee appointed by the University Grants Commission (UGC), Government of India, Srinivasa Ramanujan Centre, established by SASTRA, has been declared as an OFF-CAMPUS CENTRE under the ambit of SASTRA University. House of Ramanujan Mathematics, a museum on life and works of the

Mathematical prodigy, Srinivasa Ramanujan, also exists on this campus. SASTRA purchased the house where Srinivasa Ramanujan lived at Kumabakonam and renovated it.<sup>[76]</sup>

In 2011, on the 125th anniversary of his birth, the Indian Government declared that 22 December will be celebrated every year as *National Mathematics Day*.<sup>[77]</sup> Then Indian Prime Minister Manmohan Singh also declared that the year 2012 would be celebrated as the National Mathematics Year.<sup>[78]</sup>

In popular culture

- *The Man Who Knew Infinity: A Life of the Genius Ramanujan* (ISBN 978-0-684-19259-8) is a biography of Ramanujan, written in 1991 by Robert Kanigel and published by Washington Square Press.
- *Uncle Petros and Goldbach's Conjecture* (ISBN 978-1-55608-010-4) is a 1992 novel by Greek author Apostolos Doxiadis. As himself.
- *The Man Who Knew Infinity* is a 2015 film based on the book by Robert Kanigel. In the film, Ramanujan is portrayed by British actor Dev Patel.<sup>[79][80][81]</sup>

- Ramanujan, an Indo-British collaboration film, chronicling the life of Ramanujan, was released in 2014 by the independent film company Camphor Cinema.<sup>[82]</sup> The cast and crew include director Gnana Rajasekaran, cinematographer Sunny Joseph and editor B. Lenin.<sup>[83][84]</sup> Popular Indian and English stars Abhinay Vaddi, Suhasini Maniratnam, Bhama, Kevin McGowan and Michael Lieber star in pivotal roles.<sup>[85]</sup>
- The thriller novel The Steradian Trail by M. N. Krish weaves Ramanujan and his accidental discovery into its plot connecting religion, mathematics, finance and economics.<sup>[86][87]</sup>

- Partition, a play by Ira Hauptman about Hardy and Ramanujan, first performed in 2013.<sup>[88][89][90][91]</sup>
- A play, *First Class Man* by Alter Ego Productions,<sup>[92]</sup> was based on David Freeman's *First Class Man*. The play is centred around Ramanujan and his complex and dysfunctional relationship with Hardy. On 16 October 2011, it was announced that Roger Spottiswoode, best known for his James Bond film *Tomorrow Never Dies*, is working on the film version, starring actor Siddharth. Like the book and play it is also titled *The First Class Man*.<sup>[93]</sup>

- *A Disappearing Number* is a recent British stage production by the company Complicite that explores the relationship between Hardy and Ramanujan.<sup>[94]</sup>
- The novel *The Indian Clerk* by David Leavitt explores in fiction the events following Ramanujan's letter to Hardy.<sup>[95][96]</sup>
- Google honoured him on his 125th birth anniversary by replacing its logo with a doodle on its home page.<sup>[97][98]</sup>
- Ramanujan was mentioned in the 1997 film *Good Will Hunting*, in a scene where professor Gerald Lambeau (Stellan Skarsgard) explains to Sean

Maguire (Robin Williams) the genius of Will Hunting (Matt Damon) by comparing him to Ramanujan.<sup>[99]</sup>

- On 22 March 1988, the PBS series Nova aired a documentary about Ramanujan, "The Man Who Loved Numbers" (Season 15, Episode 19).<sup>[100]</sup>
- In the book Hyperspace by Michio Kaku, Ramanujan's contributions to Superstring theory and a brief synopsis of his life are given in *Part II Unification in Ten Dimensions* in the chapter *Superstrings* under the sections *Mystery of Modular Functions* and *Reinventing 100 Years of Mathematics*.

- The 2013 documentary *The Genius of Srinivasa Ramanujan* explores his achievements in theory of numbers. It includes interviews with number theorists like A. Raghuram and Ken Ono.<sup>[101]</sup>
- In December, 2017 the Ramanujan Math Park, a museum dedicated to mathematics education, was established in Chittoor, Andhra Pradesh, India.
- In 2010 , a college under University of Delhi<sup>[102]</sup> named Deshbandhu college (Evening) was named after Ramanujan as Ramanujan College.<sup>[103]</sup>

## Further works of

# Ramanujan's mathematics

- George E. Andrews and Bruce C. Berndt, *Ramanujan's Lost Notebook: Part I* (Springer, 2005, ISBN 0-387-25529-X)<sup>[104]</sup>
- George E. Andrews and Bruce C. Berndt, *Ramanujan's Lost Notebook: Part II*, (Springer, 2008, ISBN 978-0-387-77765-8)
- George E. Andrews and Bruce C. Berndt, *Ramanujan's Lost Notebook: Part III*, (Springer, 2012, ISBN 978-1-4614-3809-0)
- George E. Andrews and Bruce C. Berndt, *Ramanujan's Lost Notebook:*

*Part IV*, (Springer, 2013, ISBN 978-1-4614-4080-2)

- M. P. Chaudhary, *A simple solution of some integrals given by Srinivasa Ramanujan*, (Resonance: J. Sci. Education – publication of Indian Academy of Science, 2008)<sup>[105]</sup>

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This book was originally published in 1927<sup>[106]</sup> after Ramanujan's death. It contains the 37 papers published in professional journals by Ramanujan during his lifetime. The third reprint contains additional commentary by Bruce C. Berndt.

- S. Ramanujan (1957). *Notebooks (2 Volumes)*. Bombay: Tata Institute of Fundamental Research.

These books contain photocopies of the original notebooks as written by Ramanujan.

- S. Ramanujan (1988). *The Lost Notebook and Other Unpublished Papers*. New Delhi: Narosa. ISBN 978-3-540-18726-4.

This book contains photo copies of the pages of the "Lost Notebook".

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This was produced from scanned and microfilmed images of the original

manuscripts by expert archivists of Roja Muthiah Research Library, Chennai.

## See also

- [1729 \(number\)](#).
- [Brown numbers](#)
- [List of amateur mathematicians](#)
- [List of Indian mathematicians](#)
- [Ramanujan graph](#)
- [Ramanujan summation](#)
- [Ramanujan's constant](#)
- [Ramanujan's ternary quadratic form](#)
- [Rank of a partition](#)

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- Who Was Ramanujan?
- A Study Group For Mathematics:  
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